Exercise 4.

1.) SQL Query

```
SELECT    e.EmployeeID
          e.FirstName
          e.Last Name,
          e.Department,
          e.Salary,
          p.ProjectID,
          p.Project Name,
          p.Budget
          p.Status
FROM      Employees e
INNER JOIN  Projects p
ON  e.EmployeeID = p.EmployeeID
```

Result

| Employee ID | First Name | Last Name | Department | Salary | Project ID |
|---|---|---|---|---|---|
| 1 | John | Doe | IT | 70 000 | 101 |
| 1 | John | Doe | IT | 70 000 | 103 |
| 2 | Alice | Smith | HR | 60 000 | 102 |
| 3 | Bob | Johnson | Finance | 75 000 | 104 |
| 4 | David | Brown | IT | 72 000 | Null |
| 5 | Emma | Wilson | Sales | 65 000 | 105 |
| 6 | Michael | Clark | Finance | 80 000 | 106 |

| Project Name | | |
|---|---|---|
| AI Development | 100 000 | Completed |
| Cyber security Audit | 75 000 | Pending |
| Employee Training | 50 000 | Ongoing |
| Financial Analytics | 80000 | Ongoing |
| Null | Null | Null |
| Market Expansion | 6500 | Completed |

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| Project Name | | |
|---|---|---|
| AI Development | 100 000 | Completed |
| Cyber security Audit | 75 000 | Pending |
| Employee Training | 50 000 | Ongoing |
| Financial Analytics | 80000 | Ongoing |
| Null | Null | Null |
| Market Expansion | 65000 | Completed |
| Risk Management | 80000 | Pending |

a.) SQL Query

```
SELECT    e.EmployeeID
          e.FirstName,
          e.Last Name,
          e.Department,
          e.Salary,
          p.ProjectID,
          p.Project Name,
          p.Budget
          p.Status
FROM      Employees e
LEFT JOIN Projects p
ON e.EmployeeID = p.EmployeeID
```

| EmployeeID | First Name | Last Name | Department | Salary | ProjectID |
|---|---|---|---|---|---|
| 1 | John | Doe | IT | 70 000 | 101 |
| 1 | John | Doe | IT | 70 000 | 103 |
| 2 | Alice | Smith | HR | 60 000 | 102 |
| 3 | Bob | Johnson | Finance | 75 000 | 104 |

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| Employee ID | First Name | Last Name | Department | Salary | Project ID |
|---|---|---|---|---|---|
| 1 | John | Doe | IT | 70 000 | 101 |
| 1 | John | Doe | IT | 70 000 | 103 |
| 2 | Alice | Smith | HR | 60 000 | 102 |
| 3 | Bob | Johnson | Finance | 75 000 | 104 |
| 4 | David | Brown | IT | 72 000 | NULL |
| 5 | Emma | Wilson | Sales | 65 000 | 105 |
| 6 | Michael | Clark | Finance | 80 000 | 106 |

| Project Name | Budget | Status |
|---|---|---|
| AI Development | 100 000 | Completed |
| Cyber security Audit | 75 000 | Pending |
| Employee Training | 50 000 | Ongoing |
| Financial Analysis | 90000 | Ongoing |
| Null | Null | Null |
| Market Expansion | 65000 | Completed |
| Risk Management | 80000 | Pending |

B) SQL Query

```
SELECT    p.ProjectID,
          p.ProjectName,
          p.Budget,
          p.Status,
          e.EmploymentID,
          e.FirstName,
          e.LastName,
          e.Department
          e.Salary

FROM      Employees e
Right     JOIN      Projects p
ON        e.EmployeeID = p.EmployeeID
```


| Employee ID | First Name | Last Name | Department | Salary | Project ID |
|---|---|---|---|---|---|
| 1 | John | Doe | IT | 70 000 | 101 |
| 1 | John | Doe | IT | 70 000 | 103 |
| 2 | Alice | Smith | HR | 60 000 | 102 |
| 3 | Bob | Johnson | Finance | 75 000 | 104 |
| 4 | David | Brown | IT | 72 000 | NULL |
| 5 | Emma | Wilson | Sales | 65 000 | 105 |
| 6 | Michael | Clark | Finance | 80 000 | 106 |




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ON e.EmployeeID = p.EmployeeID

| EmployeeID | First Name | Last Name | Department | Salary | ProjectID |
|---|---|---|---|---|---|
| 1 | John | Doe | IT | 70 000 | 101 |
| 1 | John | Doe | IT | 70 000 | 103 |
| 2 | Alice | Smith | HR | 60 000 | 102 |
| 3 | Bob | Johnson | Finance | 75 000 | 104 |
| 4 | David | Brown | IT | 72 000 | NULL |
| 5 | Emma | Wilson | Sales | 65 000 | 105 |
| 6 | Michael | Clark | Finance | 80 000 | 106 |

| Project Name | Budget | Status |
|---|---|---|
| AI Development | 100 000 | Completed |
| Cybersecurity Audit | 75 000 | Pending |
| Employee Training | 50 000 | Ongoing |
| Financial Analysis | 90000 | Ongoing |
| Null | Null | Null |
| Market Expansion | 65000 | Completed |
| Risk Management | 80000 | Pending |

Q.4

```
SELECT    e.EmployeeID
          e.FirstName
          e.LastName,
          e.Department,
          e.Salary,
          p.ProjectID,
          p.ProjectName,
          p.Budget
          p.Status
FROM      Employees e
Full Outer JOIN  Projects p
ON  e.EmployeeID = p.EmployeeID
```

| EmployeeID | First Name | Last Name | Department | Salary | ProjectID |
|---|---|---|---|---|---|
| 1 | John | Doe | IT | 70 000 | 101 |
| 1 | John | Doe | IT | 70 000 | 103 |

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| Employee ID | First Name | Last Name | Department | Salary | Project ID |
|---|---|---|---|---|---|
| 1 | John | Doe | IT | 70 000 | 101 |
| 1 | John | Doe | IT | 70 000 | 103 |
| 2 | Alice | Smith | HR | 60 000 | 102 |
| 3 | Bob | Johnson | Finance | 75 000 | 104 |
| 4 | David | Brown | IT | 72 000 | NULL |
| 5 | Emma | Wilson | Sales | 65 000 | 105 |
| 6 | Michael | Clark | Finance | 80 000 | 106 |

| Project Name | Budget | Status |
|---|---|---|
| AI Development | 100 000 | Completed |
| Cyber security Audit | 75 000 | Pending |
| Employee Training | 50 000 | Ongoing |
| Financial Analysis | 90000 | Ongoing |
| Null | Null | Null |
| Market Expansion | 65000 | Completed |
| Risk Management | 80000 | Pending |

## Question 7

### SQL Query

```
SELECT  Employee ID, First Name, Last Name,
        Department, Salary
FROM    Employees
WHERE   Salary > 70 000;
```

Result

| Employee ID | First Name | Last Name | Department | Salary |
|---|---|---|---|---|
| 3 | Bob | Johnson | Finance | 75000 |
| 4 | David | Brown | IT | 72000 |
| 6 | Michael | Clark | Finance | 80000 |

Q 8



SQL Query

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| | | | | |
|---|---|---|---|---|
| 4 | David | Brown | IT | 72000 |
| 6 | Michael | Clark | Finance | 80000 |

Q8
SQL Query

```
SELECT  EmployeeID, FirstName, LastName, Department,
        Salary
FROM    Employees
WHERE   Department IN('IT, 'Finance');
```

| Employee ID | First Name | Last Name | Department | Salary |
|---|---|---|---|---|
| 1 | John | Doe | IT | 70 000 |
| 3 | Bob | Johnson | Finance | 75 000 |
| 4 | David | Brown | IT | 72 000 |
| 6 | Michael | Clark | Finance | 80 000 |

Q9
SQL Query

```
SELECT  ProjectID, ProjectName, Budget, Status
FROM    Projects
WHERE   Status <> 'Completed';
```

Result

| Project ID | Project Name | Budget | Status |
|---|---|---|---|
| 102 | Employee Training | 50 000 | Ongoing |
| 103 | Cybersecurity Audit | 75 000 | Pending |
| 104 | Financial Analysis | 90 000 | Ongoing |
| 106 | Risk Management | 50 000 | Pending |

Question 10
SQL Query

```
SELECT  ProjectID, ProjectName, Budget, Status
FROM    Projects
WHERE   Budget > 70 000
```

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Question 10
SQL Query

```
SELECT  ProjectID, ProjectName, Budget, Status
FROM    Projects
WHERE   Budget > 70,000
AND     Status <> 'Completed';
```

Result

| Project ID | ProjectName | Budget | Status |
|---|---|---|---|
| 103 | Cybersecurity Audit | 75000 | Pending |
| 104 | Financial Analyst | 90000 | Ongoing |
| 106 | Risk Management | 80000 | Pending |

Q11
SQL query

```
SELECT  EmployeeID, FirstName, LastName, Department,
        Salary, City
FROM    employees
WHERE   City IN ('New York', 'Toronto')
ORDER By Salary DESC;
```

Result

| Employee ID | FirstName | LastName | Department | Salary | City |
|---|---|---|---|---|---|
| 6 | Michael | Clark | finance | 80000 | New York |
| 3 | Bob | Johnson | Finance | 75000 | Toronto |
| 1 | John | Doe | IT | 70000 | New York |

Q12
SQL Query

```
SELECT  EmployeeID, FirstName, Last Name, Department,
        Salary
FROM
```

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| | | | | | |
|---|---|---|---|---|---|
| | John | Doe | IT | 70000 | New York |

Q 12
SQL Query

```
SELECT  EmployeeID, FirstName, LastName, Department,
        Salary
FROM    Employees
ORDER BY  Salary DESC
LIMIT 3;
```

Result

| EmployeeID | FirstName | LastName | Department | Salary |
|---|---|---|---|---|
| 6 | Michael | Clark | Finance | 80 000 |
| 3 | Bob | Johnson | Finance | 75 000 |
| 4 | David | Brown | IT | 72 000 |

Q 13
SQL Query

```
SELECT  Department, SUM(Salary) AS TotalSalary
FROM    Employees
GROUP BY  Department
ORDER BY  TotalSalary DESC;
```

Result

| Department | Total Salary |
|---|---|
| Finance | 155 000 |
| IT | 142 000 |
| Sales | 65 000 |
| HR | 60 000 |

Q 14
SQL Query

```
SELECT  Cit, AVG(Salary) AS AverageSalary
FROM    Employees
```



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| | |
|---|---|
| R | 60 000 |

Q 14

SQL Query

```
SELECT City, AVG(Salary) AS AverageSalary
FROM Employees
GROUP BY City
HAVING AVG(Salary) > 65 000;
```

Result

| City | Average Salary |
|---|---|
| New York | 75000 |
| Toronto | 75000 |
| London | 72 000 |

Q 15

SQL Query

```
SELECT Department, COUNT(EmployeeID) As EmployeeCount
FROM Employees
GROUP BY Department
HAVING COUNT(EmployeeID) >1 ;
```

Result

| Department | EmployeeCount |
|---|---|
| IT | 2 |
| Finance | 2 |

Q 16

SQL Query

```
SELECT Status, COUNT(ProjectID) As ProjectCount
FROM Projects
GROUP BY Status
HAVING COUNT(ProjectID)>=2;
```

Result

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Q16

SQL Query

```
SELECT  Status, COUNT (ProjectID) As Project Count
FROM    Projects
GROUP   BY  Status
HAVING  COUNT (ProjectID)>=2;
```

Result

| Status | Project Count |
|---|---|
| Completed | 2 |
| Ongoing | 2 |
| Pending | 2 |

Q17

SQL Query

```
SELECT    e. Employee ID
          e. FirstName
          e. LastName
  SUM (p. Budget)  AS  TotalProjectBudget
FROM    Employee e
JOIN    Projects p
ON      e LastName
HAVING  SUM.p(.Budget)>15000
```

Result

| employee ID | FirstName | LastName | TotalProjectBudge |
|---|---|---|---|
| 1 | John | Doe | 175000 |

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